Momentum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate and compare signed momentum.

SUCCESS CRITERIA - I can...

1. Multiply mass by velocity with correct units.
2. Include direction or a sign.
3. Distinguish zero total momentum from no motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

An asteroid's crossing window is days 13-17; Earth's is 15-19. Find the overlap and explain why this is not proof of a collision.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p = mv$; $p_{total} = p_1 + p_2$.
Use signed velocities; momentum units are kg·m/s.

Worked reasoning: A 3 kg cart moving left at 2 m/s has momentum -6 kg·m/s when right is positive. The minus sign gives direction, not a negative mass.

Your turn: Draw separate velocity and momentum arrows, labeling their different units.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and interpret the model

The two carts have the same speed in opposite directions. Which momentum values are positive if right is positive?



2 kg; 3 m/s right

2 kg; 3 m/s left

Marzano check - circle: ① ② ③

04 Apply the physics

A 0.50 kg cart moves at +6 m/s; a 1.5 kg cart moves at -2 m/s. Find each momentum and total momentum. Draw the momentum arrows and explain why zero total does not mean both carts are at rest.

Object / trial	Mass	Velocity	Momentum
A	___	___	___
B	___	___	___

Marzano check - circle: ① ② ③

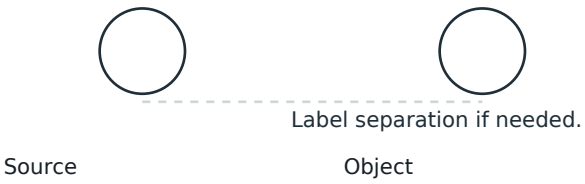
05 Explain your reasoning

One cart doubles mass at the same velocity. Another doubles speed at the same mass. Compare the momentum multipliers.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Outside a spherical source, field at r is 144 N/kg. Find field at 6 r and the force on 0.8 kg there.
Label the masses, field or force relevant to the question.



Marzano check - circle: 1 2 3

07 Exit check

A cart has momentum -12 kg·m/s and mass 3 kg. Find velocity. Then calculate its momentum if velocity reverses without changing speed.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Three carts have masses 1, 2, 3 kg and velocities +4,-5, v m/s. Choose v for zero total momentum. Then reverse only the 1 kg cart and recalculate the total.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3